Experiment 6

Implementing N-GRAM model with (NLTK / spaCy / Stanford) (ENGLISH) in Python

NLTK — Bi-Gram, Tri-Gram & Four-Gram

```
# Install NLTK if required
!pip install nltk
import nltk
from nltk.util import ngrams
from collections import Counter
# Download tokenizer
nltk.download('punkt')
nltk.download('punkt_tab')

# ------------------------------------------------
# 1. CORPUS - 12 SENTENCES
# ------------------------------------------------

corpus = [
    "I like to study AI",
    "I like to study NLP",
    "I like AI projects",
    "AI is a branch of computer science",
    "NLP is a subdomain of AI",
    "NLP is useful for language processing",
    "Students like to study AI",
    "Students like NLP projects",
    "AI projects are interesting",
    "NLP projects are useful",
    "I study computer science",
    "I like computer science"
]

# Convert sentences into words
tokens = []

for sentence in corpus:
    words = nltk.word_tokenize(sentence.lower())
    tokens.extend(words)

# ------------------------------------------------
# 2. UNIGRAM / WORD FREQUENCY TABLE
# ------------------------------------------------

unigram = Counter(tokens)

print("\n========== WORD FREQUENCY TABLE ==========")
print("Word\t\tFrequency")
print("----------------------------------------")

for word, freq in unigram.most_common():
```

```
    print(f"{word:<15}{freq}")

# ------------------------------------------------
# 3. BIGRAM
# ------------------------------------------------

bigram = list(ngrams(tokens, 2))
bigram_freq = Counter(bigram)

print("\n========== BIGRAM FREQUENCY TABLE ==========")
print("Bigram\t\t\tFrequency")
print("-----------------------------------------")

for bg, freq in bigram_freq.most_common():
    print(f"{' '.join(bg):<25}{freq}")

# ------------------------------------------------
# 4. TRIGRAM
# ------------------------------------------------

trigram = list(ngrams(tokens, 3))
trigram_freq = Counter(trigram)

print("\n========== TRIGRAM FREQUENCY TABLE ==========")
print("Trigram\t\t\t\tFrequency")
print("-------------------------------------------")

for tg, freq in trigram_freq.most_common():
    print(f"{' '.join(tg):<30}{freq}")

# ------------------------------------------------
# 5. FOUR-GRAM
# ------------------------------------------------

fourgram = list(ngrams(tokens, 4))
fourgram_freq = Counter(fourgram)

print("\n========== FOUR-GRAM FREQUENCY TABLE ==========")
print("Four-gram\t\t\t\tFrequency")
print("-------------------------------------------")

for fg, freq in fourgram_freq.most_common():
    print(f"{' '.join(fg):<35}{freq}")

# ------------------------------------------------
# 6. NEXT WORD PREDICTION
# ------------------------------------------------

def predict_next_word(context, ngram_freq, n):
    context = tuple(context.lower().split())
```

```
    candidates = {}

    for gram, freq in ngram_freq.items():

        if gram[:-1] == context:
            candidates[gram[-1]] = freq

    if candidates:
        predicted_word = max(candidates, key=candidates.get)
        return predicted_word

    return "No prediction found"


# ------------------------------------------------
# BIGRAM PREDICTION
# ------------------------------------------------

print("\n========== NEXT WORD PREDICTION ==========")

context = "i like"

# For bigram, use last word
bigram_context = context.split()[-1]

prediction = predict_next_word(
    bigram_context,
    bigram_freq,
    2
)

print("\nBigram:")
print("Input : I like")
print("Predicted next word :", prediction)


# ------------------------------------------------
# TRIGRAM PREDICTION
# ------------------------------------------------

trigram_context = "i like"

prediction = predict_next_word(
    trigram_context,
    trigram_freq,
    3
)

print("\nTrigram:")
```

```
print("Input : I like")
print("Predicted next word :", prediction)


# ------------------------------------------------
# FOUR-GRAM PREDICTION
# ------------------------------------------------

fourgram_context = "i like to"

prediction = predict_next_word(
    fourgram_context,
    fourgram_freq,
    4
)

print("\nFour-gram:")
print("Input : I like to")
print("Predicted next word :", prediction)
```

Output

```
========== WORD FREQUENCY TABLE ==========
Word            Frequency
-----------------------------------------
like            6
ai              6
i               5
nlp             5
study           4
projects        4
to              3
is              3
computer        3
science         3
a               2
of              2
useful          2
students        2
are             2
branch          1
subdomain       1
for             1
language        1
processing      1
interesting     1

========== BIGRAM FREQUENCY TABLE ==========
Bigram                  Frequency
-------------------------------------------
i like                    4
like to                   3
to study                  3
computer science          3
study ai                  2
```

```
ai projects              2
projects ai              2
is a                     2
nlp is                   2
students like            2
nlp projects             2
projects are             2
ai i                     1
study nlp                1
nlp i                    1
like ai                  1
ai is                    1
a branch                 1
branch of                1
of computer              1
science nlp              1
a subdomain              1
subdomain of             1
of ai                    1
ai nlp                   1
is useful                1
useful for               1
for language             1
language processing      1
processing students      1
ai students              1
like nlp                 1
are interesting          1
interesting nlp          1
are useful               1
useful i                 1
i study                  1
study computer           1
science i                1
like computer            1

========== TRIGRAM FREQUENCY TABLE ==========
Trigram                         Frequency
-----------------------------------------------
like to study                   3
i like to                       2
to study ai                     2
study ai i                      1
ai i like                       1
to study nlp                    1
study nlp i                     1
nlp i like                      1
i like ai                       1
like ai projects                1
ai projects ai                  1
projects ai is                  1
ai is a                         1
is a branch                     1
a branch of                     1
branch of computer              1
of computer science             1
computer science nlp            1
science nlp is                  1
nlp is a                        1
is a subdomain                  1
a subdomain of                  1
```

```
subdomain of ai               1
of ai nlp                     1
ai nlp is                     1
nlp is useful                 1
is useful for                 1
useful for language           1
for language processing       1
language processing students  1
processing students like      1
students like to              1
study ai students             1
ai students like              1
students like nlp             1
like nlp projects             1
nlp projects ai               1
projects ai projects          1
ai projects are               1
projects are interesting      1
are interesting nlp           1
interesting nlp projects      1
nlp projects are              1
projects are useful           1
are useful i                  1
useful i study                1
i study computer              1
study computer science        1
computer science i            1
science i like                1
i like computer               1
like computer science         1

========== FOUR-GRAM FREQUENCY TABLE ==========
Four-gram                                Frequency
-----------------------------------------------
i like to study                          2
like to study ai                         2
to study ai i                            1
study ai i like                          1
ai i like to                             1
like to study nlp                        1
to study nlp i                           1
study nlp i like                         1
nlp i like ai                            1
i like ai projects                       1
like ai projects ai                      1
ai projects ai is                        1
projects ai is a                         1
ai is a branch                           1
is a branch of                           1
a branch of computer                     1
branch of computer science               1
of computer science nlp                  1
computer science nlp is                  1
science nlp is a                         1
nlp is a subdomain                       1
is a subdomain of                        1
a subdomain of ai                        1
subdomain of ai nlp                      1
of ai nlp is                             1
ai nlp is useful                         1
nlp is useful for                        1
```

```
is useful for language             1
useful for language processing     1
for language processing students   1
language processing students like  1
processing students like to        1
students like to study             1
to study ai students               1
study ai students like             1
ai students like nlp               1
students like nlp projects         1
like nlp projects ai               1
nlp projects ai projects           1
projects ai projects are           1
ai projects are interesting        1
projects are interesting nlp       1
are interesting nlp projects       1
interesting nlp projects are       1
nlp projects are useful            1
projects are useful i              1
are useful i study                 1
useful i study computer            1
i study computer science           1
study computer science i           1
computer science i like            1
science i like computer            1
i like computer science            1
========== NEXT WORD PREDICTION ==========

Bigram:
Input : I like
Predicted next word : to

Trigram:
Input : I like
Predicted next word : to

Four-gram:
Input : I like to
Predicted next word : study
```

SpaCY — Bi-Gram, Tri-Gram & Four-Gram

```
!pip install spacy
!python -m spacy download en_core_web_sm
   return [tuple(tokens[i:i+n]) for i in range(len(tokens)-n+1)]
import spacy
from collections import Counter

# Load spaCy English model
nlp = spacy.load("en_core_web_sm")


# ------------------------------------------------
# 1. CORPUS - 12 SENTENCES
# ------------------------------------------------

corpus = [
```

```
    "I like to study AI",
    "I like to study NLP",
    "I like AI projects",
    "AI is a branch of computer science",
    "NLP is a subdomain of AI",
    "NLP is useful for language processing",
    "Students like to study AI",
    "Students like NLP projects",
    "AI projects are interesting",
    "NLP projects are useful",
    "I study computer science",
    "I like computer science"
]


# ------------------------------------------------
# 2. TOKENIZATION USING SPACY
# ------------------------------------------------

tokens = []

for sentence in corpus:
    doc = nlp(sentence.lower())

    for token in doc:
        if not token.is_space:
            tokens.append(token.text)

print("\n========== TOKENS ==========")
print(tokens)


# ------------------------------------------------
# 3. UNIGRAM
# ------------------------------------------------

unigram = Counter(tokens)

print("\n========== WORD FREQUENCY TABLE ==========")
print("Word\t\tFrequency")
print("----------------------------------------")

for word, freq in unigram.most_common():
    print(f"{word:<15}{freq}")


# ------------------------------------------------
# FUNCTION TO CREATE N-GRAMS
# ------------------------------------------------
```

```
def make_ngrams(tokens, n):
    return [
        tuple(tokens[i:i+n])
        for i in range(len(tokens) - n + 1)
    ]


# ------------------------------------------------
# 4. BIGRAM
# ------------------------------------------------

bigram = make_ngrams(tokens, 2)
bigram_freq = Counter(bigram)

print("\n========== BIGRAM FREQUENCY TABLE ==========")
print("Bigram\t\t\tFrequency")
print("-----------------------------------------")

for bg, freq in bigram_freq.most_common():
    print(f"{' '.join(bg):<25}{freq}")


# ------------------------------------------------
# 5. TRIGRAM
# ------------------------------------------------

trigram = make_ngrams(tokens, 3)
trigram_freq = Counter(trigram)

print("\n========== TRIGRAM FREQUENCY TABLE ==========")
print("Trigram\t\t\t\tFrequency")
print("-----------------------------------------")

for tg, freq in trigram_freq.most_common():
    print(f"{' '.join(tg):<30}{freq}")


# ------------------------------------------------
# 6. FOUR-GRAM
# ------------------------------------------------

fourgram = make_ngrams(tokens, 4)
fourgram_freq = Counter(fourgram)

print("\n========== FOUR-GRAM FREQUENCY TABLE ==========")
print("Four-gram\t\t\t\tFrequency")
print("-----------------------------------------")

for fg, freq in fourgram_freq.most_common():
    print(f"{' '.join(fg):<35}{freq}")
```

```
# ------------------------------------------------
# 7. NEXT WORD PREDICTION
# ------------------------------------------------

def predict_next_word(context, ngram_freq):

    context = tuple(context.lower().split())

    candidates = {}

    for gram, freq in ngram_freq.items():

        if gram[:-1] == context:
            candidates[gram[-1]] = freq

    if candidates:
        return max(candidates, key=candidates.get)

    return "No prediction found"


# ------------------------------------------------
# 8. PREDICTION
# ------------------------------------------------

print("\n========== NEXT WORD PREDICTION ==========")


# BIGRAM
prediction = predict_next_word(
    "like",
    bigram_freq
)

print("\nBigram:")
print("Input : I like")
print("Predicted next word :", prediction)


# TRIGRAM
prediction = predict_next_word(
    "i like",
    trigram_freq
)

print("\nTrigram:")
print("Input : I like")
print("Predicted next word :", prediction)
```

```
# FOUR-GRAM
prediction = predict_next_word(
   "i like to",
   fourgram_freq
)

print("\nFour-gram:")
print("Input : I like to")
print("Predicted next word :", prediction)
```

Output

```
========== WORD FREQUENCY TABLE ==========
Word            Frequency
-----------------------------------------
like            6
ai              6
i               5
nlp             5
study           4
projects        4
to              3
is              3
computer        3
science         3
a               2
of              2
useful          2
students        2
are             2
branch          1
subdomain       1
for             1
language        1
processing      1
interesting     1


========== BIGRAM FREQUENCY TABLE ==========
Bigram                  Frequency
-------------------------------------------
i like                    4
like to                   3
to study                  3
computer science          3
study ai                  2
ai projects               2
projects ai               2
is a                      2
nlp is                    2
students like             2
nlp projects              2
projects are              2
ai i                      1
study nlp                 1
```

nlp i	1
like ai	1
ai is	1
a branch	1
branch of	1
of computer	1
science nlp	1
a subdomain	1
subdomain of	1
of ai	1
ai nlp	1
is useful	1
useful for	1
for language	1
language processing	1
processing students	1
ai students	1
like nlp	1
are interesting	1
interesting nlp	1
are useful	1
useful i	1
i study	1
study computer	1
science i	1
like computer	1

========== TRIGRAM FREQUENCY TABLE ==========

Trigram	Frequency
like to study	3
i like to	2
to study ai	2
study ai i	1
ai i like	1
to study nlp	1
study nlp i	1
nlp i like	1
i like ai	1
like ai projects	1
ai projects ai	1
projects ai is	1
ai is a	1
is a branch	1
a branch of	1
branch of computer	1
of computer science	1
computer science nlp	1
science nlp is	1
nlp is a	1
is a subdomain	1
a subdomain of	1
subdomain of ai	1
of ai nlp	1
ai nlp is	1
nlp is useful	1
is useful for	1
useful for language	1
for language processing	1
language processing students	1
processing students like	1

```
students like to              1
study ai students             1
ai students like              1
students like nlp             1
like nlp projects             1
nlp projects ai               1
projects ai projects          1
ai projects are               1
projects are interesting      1
are interesting nlp           1
interesting nlp projects      1
nlp projects are              1
projects are useful           1
are useful i                  1
useful i study                1
i study computer              1
study computer science        1
computer science i            1
science i like                1
i like computer               1
like computer science         1

========== FOUR-GRAM FREQUENCY TABLE ==========
Four-gram                                Frequency
------------------------------------------------
i like to study                        2
like to study ai                       2
to study ai i                          1
study ai i like                        1
ai i like to                           1
like to study nlp                      1
to study nlp i                         1
study nlp i like                       1
nlp i like ai                          1
i like ai projects                     1
like ai projects ai                    1
ai projects ai is                      1
projects ai is a                       1
ai is a branch                         1
is a branch of                         1
a branch of computer                   1
branch of computer science             1
of computer science nlp                1
computer science nlp is                1
science nlp is a                       1
nlp is a subdomain                     1
is a subdomain of                      1
a subdomain of ai                      1
subdomain of ai nlp                    1
of ai nlp is                           1
ai nlp is useful                       1
nlp is useful for                      1
is useful for language                 1
useful for language processing         1
for language processing students       1
language processing students like      1
processing students like to            1
students like to study                 1
to study ai students                   1
study ai students like                 1
ai students like nlp                   1
```

```
students like nlp projects         1
like nlp projects ai               1
nlp projects ai projects           1
projects ai projects are           1
ai projects are interesting        1
projects are interesting nlp       1
are interesting nlp projects       1
interesting nlp projects are       1
nlp projects are useful            1
projects are useful i              1
are useful i study                 1
useful i study computer            1
i study computer science           1
study computer science i           1
computer science i like            1
science i like computer            1
i like computer science            1

========== NEXT WORD PREDICTION ==========

Bigram:
Input : I like
Predicted next word : to

Trigram:
Input : I like
Predicted next word : to

Four-gram:
Input : I like to
Predicted next word : study
```

Stanza — Bi-Gram, Tri-Gram & Four-Gram

```
!pip install stanza
import stanza
from collections import Counter

# Download English model
stanza.download('en')

# Create Stanford/Stanza pipeline
nlp = stanza.Pipeline(
    lang='en',
    processors='tokenize'
)

# ------------------------------------------------
# 1. CORPUS - 12 SENTENCES
# ------------------------------------------------

corpus = [
    "I like to study AI",
    "I like to study NLP",
    "I like AI projects",
    "AI is a branch of computer science",
```

```
    "NLP is a subdomain of AI",
    "NLP is useful for language processing",
    "Students like to study AI",
    "Students like NLP projects",
    "AI projects are interesting",
    "NLP projects are useful",
    "I study computer science",
    "I like computer science"
]


# ------------------------------------------------
# 2. TOKENIZATION USING STANFORD NLP
# ------------------------------------------------

tokens = []

for sentence in corpus:

    doc = nlp(sentence.lower())

    for sent in doc.sentences:
        for word in sent.words:
            tokens.append(word.text)


print("\n========== TOKENS ==========")
print(tokens)


# ------------------------------------------------
# 3. UNIGRAM
# ------------------------------------------------

unigram = Counter(tokens)

print("\n========== WORD FREQUENCY TABLE ==========")
print("Word\t\tFrequency")
print("-----------------------------------------")

for word, freq in unigram.most_common():
    print(f"{word:<15}{freq}")


# ------------------------------------------------
# FUNCTION TO CREATE N-GRAMS
# ------------------------------------------------

def make_ngrams(tokens, n):
```

```
    result = []

    for i in range(len(tokens) - n + 1):
        result.append(tuple(tokens[i:i+n]))

    return result


# ------------------------------------------------
# 4. BIGRAM
# ------------------------------------------------

bigram = make_ngrams(tokens, 2)
bigram_freq = Counter(bigram)

print("\n========== BIGRAM FREQUENCY TABLE ==========")
print("Bigram\t\t\tFrequency")
print("-------------------------------------------")

for bg, freq in bigram_freq.most_common():
    print(f"{' '.join(bg):<25}{freq}")


# ------------------------------------------------
# 5. TRIGRAM
# ------------------------------------------------

trigram = make_ngrams(tokens, 3)
trigram_freq = Counter(trigram)

print("\n========== TRIGRAM FREQUENCY TABLE ==========")
print("Trigram\t\t\t\tFrequency")
print("-------------------------------------------")

for tg, freq in trigram_freq.most_common():
    print(f"{' '.join(tg):<30}{freq}")


# ------------------------------------------------
# 6. FOUR-GRAM
# ------------------------------------------------

fourgram = make_ngrams(tokens, 4)
fourgram_freq = Counter(fourgram)

print("\n========== FOUR-GRAM FREQUENCY TABLE ==========")
print("Four-gram\t\t\t\tFrequency")
print("-------------------------------------------")

for fg, freq in fourgram_freq.most_common():
```

```
    print(f"{' '.join(fg):<35}{freq}")


# ------------------------------------------------
# 7. NEXT WORD PREDICTION
# ------------------------------------------------

def predict_next_word(context, ngram_freq):

    context = tuple(context.lower().split())

    candidates = {}

    for gram, freq in ngram_freq.items():

        if gram[:-1] == context:
            candidates[gram[-1]] = freq

    if candidates:
        return max(candidates, key=candidates.get)

    return "No prediction found"


# ------------------------------------------------
# 8. NEXT WORD PREDICTION
# ------------------------------------------------

print("\n========== NEXT WORD PREDICTION ==========")


# BIGRAM
prediction = predict_next_word(
    "like",
    bigram_freq
)

print("\nBigram:")
print("Input : I like")
print("Predicted next word :", prediction)


# TRIGRAM
prediction = predict_next_word(
    "i like",
    trigram_freq
)

print("\nTrigram:")
print("Input : I like")
```

```
print("Predicted next word :", prediction)


# FOUR-GRAM
prediction = predict_next_word(
    "i like to",
    fourgram_freq
)

print("\nFour-gram:")
print("Input : I like to")
print("Predicted next word :", prediction)
```

Output

```
========== WORD FREQUENCY TABLE ==========
Word            Frequency
------------------------------------------
like            6
ai              6
i               5
nlp             5
study           4
projects        4
to              3
is              3
computer        3
science         3
a               2
of              2
useful          2
students        2
are             2
branch          1
subdomain       1
for             1
language        1
processing      1
interesting     1

========== BIGRAM FREQUENCY TABLE ==========
Bigram                  Frequency
--------------------------------------------
i like                    4
like to                   3
to study                  3
computer science          3
study ai                  2
ai projects               2
projects ai               2
is a                      2
nlp is                    2
students like             2
nlp projects              2
projects are              2
ai i                      1
study nlp                 1
nlp i                     1
```

```
like ai                   1
ai is                     1
a branch                  1
branch of                 1
of computer               1
science nlp               1
a subdomain               1
subdomain of              1
of ai                     1
ai nlp                    1
is useful                 1
useful for                1
for language              1
language processing       1
processing students       1
ai students               1
like nlp                  1
are interesting           1
interesting nlp           1
are useful                1
useful i                  1
i study                   1
study computer            1
science i                 1
like computer             1

========== TRIGRAM FREQUENCY TABLE ==========
Trigram                           Frequency
-----------------------------------------------
like to study                    3
i like to                        2
to study ai                      2
study ai i                       1
ai i like                        1
to study nlp                     1
study nlp i                      1
nlp i like                       1
i like ai                        1
like ai projects                 1
ai projects ai                   1
projects ai is                   1
ai is a                          1
is a branch                      1
a branch of                      1
branch of computer               1
of computer science              1
computer science nlp             1
science nlp is                   1
nlp is a                         1
is a subdomain                   1
a subdomain of                   1
subdomain of ai                  1
of ai nlp                        1
ai nlp is                        1
nlp is useful                    1
is useful for                    1
useful for language              1
for language processing          1
language processing students     1
processing students like         1
students like to                 1
```

study ai students	1
ai students like	1
students like nlp	1
like nlp projects	1
nlp projects ai	1
projects ai projects	1
ai projects are	1
projects are interesting	1
are interesting nlp	1
interesting nlp projects	1
nlp projects are	1
projects are useful	1
are useful i	1
useful i study	1
i study computer	1
study computer science	1
computer science i	1
science i like	1
i like computer	1
like computer science	1

========== FOUR-GRAM FREQUENCY TABLE ==========

Four-gram	Frequency
i like to study	2
like to study ai	2
to study ai i	1
study ai i like	1
ai i like to	1
like to study nlp	1
to study nlp i	1
study nlp i like	1
nlp i like ai	1
i like ai projects	1
like ai projects ai	1
ai projects ai is	1
projects ai is a	1
ai is a branch	1
is a branch of	1
a branch of computer	1
branch of computer science	1
of computer science nlp	1
computer science nlp is	1
science nlp is a	1
nlp is a subdomain	1
is a subdomain of	1
a subdomain of ai	1
subdomain of ai nlp	1
of ai nlp is	1
ai nlp is useful	1
nlp is useful for	1
is useful for language	1
useful for language processing	1
for language processing students	1
language processing students like	1
processing students like to	1
students like to study	1
to study ai students	1
study ai students like	1
ai students like nlp	1
students like nlp projects	1

```
like nlp projects ai                1
nlp projects ai projects            1
projects ai projects are            1
ai projects are interesting         1
projects are interesting nlp        1
are interesting nlp projects        1
interesting nlp projects are        1
nlp projects are useful             1
projects are useful i               1
are useful i study                  1
useful i study computer             1
i study computer science            1
study computer science i            1
computer science i like             1
science i like computer             1
i like computer science             1

========== NEXT WORD PREDICTION ==========

Bigram:
Input : I like
Predicted next word : to

Trigram:
Input : I like
Predicted next word : to

Four-gram:
Input : I like to
Predicted next word : study
```